3.1 Unit description

Concept approach

This unit covers further mechanics, electric and magnetic fields and particle physics. The unit may be taught using either a concept approach or a context approach. The concept approach begins with a study of the laws, theories and models of physics and then explores their practical applications. This section of the specification is presented in a format for teachers who wish to use the concept approach.

Context approach

This unit is presented in a different format on page 61 for teachers who wish to use a context approach. The context approach begins with the consideration of an application that draws on many different areas of physics, and then the laws, theories and models of physics that apply to this application are studied. The context approach for this unit uses two contexts for teaching: transport and communications. Particle physics may be studied via the acceleration and detection of high-energy particles and the interpretation of experiments.

How Science Works

How Science Works – Appendix 3 should be integrated with the teaching and learning of this unit.

It is expected that students will be given opportunities to use spreadsheets and computer models to analyse and present data, and to make predictions while studying this unit.

The word ‘investigate’ indicates where students should develop their practical skills for *How Science Works*, numbers 1–6 as detailed in *Appendix 3*.

Students should communicate the outcomes of their investigations using appropriate scientific, technical and mathematical language, conventions and symbols.

Applications of physics should be studied using a range of contemporary contexts that relate to this unit.